

SUMMARY OF PRODUCT CHARACTERISTICS

1. NAME OF THE MEDICINAL PRODUCT

Aktiprol 400 mg tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Aktiprol 400 mg tablets: Each tablet contains 400 mg of amisulpride.

Excipient with known effect: Lactose monohydrate.

Each 400 mg tablet contains 200.00 mg lactose monohydrate.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Tablet.

Aktiprol 400 mg tablets: White, biconvex, capsule shaped tablets, scored on both sides, with dimensions 19 x 10 mm. The score line is only to facilitate breaking for ease of swallowing and not to divide into equal doses.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Treatment of acute or chronic schizophrenic disorders, in which positive symptoms (such as delusions, hallucinations, thought disorders) and/or negative symptoms (such as blunted emotions, emotional and social withdrawal) are pronounced, including patients with predominantly negative symptoms.

4.2 Posology and method of administration

Usually:

- if the daily dose is ≤ 400 mg, it is to be administered as a once-daily dose;
- if the daily dose exceeds 400 mg, it is to be administered as two divided doses.

Predominantly negative episodes:

Doses between 50 mg/day and 300 mg/day are recommended. Doses should be adjusted individually. The optimum dosage is about 100 mg/day.

Acute psychotic episodes

It is possible to start via the IM route for a few days, at a maximum dose of 400 mg/day, switching thereafter to oral treatment,

Oral doses between 400mg/day and 800mg/day are recommended. The maximum dose should never exceed 1200mg. Given that there has been no large-scale safety assessment of doses higher than 1200mg/day, these doses should not be used.

The dosage should be maintained or adjusted according to the patient's individual response.

In all cases, the maintenance treatment should be established individually with the minimum effective dose.

Children and adolescents

The efficacy and safety of amisulpride from the age of puberty up to 18 years has not been elucidated: available data on the use of amisulpride in adolescents with schizophrenia are limited. As a result, amisulpride is not recommended in patients from the age of puberty up to 18 years. Amisulpride is contraindicated for children under 15 years of age.

Elderly subjects

Amisulpride should be used with particular caution in this patient population due to the risk of hypotension and sedation. Dose reduction may also be required in patients with kidney failure.

Renal insufficiency

Amisulpride is eliminated via the renal route. In patients with renal insufficiency, the dose should be reduced by half when creatinine clearance (CrCl) is between 30-60 ml/min and to a third in patients with CrCl between 10-30 ml/min. As no data on patients with serious renal insufficiency (CrCl <10 ml/min) are available, careful monitoring is recommended in this population.

Hepatic insufficiency

Since amisulpride is weakly metabolized, a dosage reduction is not necessary in patients with hepatic insufficiency.

Method of administration

For oral administration.

4.3 Contraindications

- Hypersensitivity to the active ingredient or to any of the other excipients listed in section 6.1
- Concomitant prolactin-dependent tumours (e.g. pituitary gland prolactinomas or breast cancer)
- Pheochromocytoma
- Children before the onset of puberty (see section 4.2)
- Congenital prolongation of the QT interval
- Lactation (see section 4.6)
- Combination with levodopa (see section 4.5)
- Concomitant treatment with drugs that may prolong the QT interval
- Concurrent treatment with the following medications that can induce torsade de pointes:
 - Antiarrhythmics class Ia, such as quinidine and disopyramide.
 - Class III antiarrhythmic drugs such as amiodarone and sotalol.
 - Other medications such as bepridil, cisapride, sultopride, thioridazine, methadone, erythromycin IV, vincamin IV, halofantrine, pentamidine, sparfloxacin (see section 4.5).

4.4 Special warnings and precautions for use

Neuroleptic malignant syndrome

As with other neuroleptics, Neuroleptic Malignant Syndrome, a potentially fatal complication, characterized by hyperthermia, muscle rigidity, autonomic instability, altered consciousness and elevated creatine phosphokinase (CPK), may occur. In the event of hyperthermia, particularly with high daily doses, all antipsychotic drugs including Aktiprol should be discontinued.

Hyperglycaemia and diabetes mellitus

Hyperglycaemia in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients treated with atypical antipsychotics. Assessment of the relationship between atypical antipsychotics use and glucose abnormalities is complicated by the possibility of an increased background risk of diabetes mellitus in patients with schizophrenia and the increasing incidence of diabetes mellitus in the general population. Given this confounders, the relationship between atypical antipsychotic use and hyperglycaemia-related adverse events is not completely understood. However, epidemiological studies suggest an increased risk of treatment-emergent hyperglycaemia-related adverse events in patients treated with the atypical antipsychotics. Precise risk estimates for hyperglycaemia-related adverse events in patients treated with atypical antipsychotics are not available.

Patients with an established diagnosis of diabetes mellitus who are started on atypical antipsychotics should be monitored regularly for worsening of glucose control. Patients with risk factors for diabetes mellitus (e.g. obesity, family history of diabetes) who are starting treatment with atypical antipsychotics should undergo fasting blood glucose testing at the beginning of treatment and periodically during treatment. Any patient treated with atypical antipsychotics should be monitored for symptoms of hyperglycaemia including polydipsia, polyuria, polyphagia, and weakness. Patients who develop symptoms of hyperglycaemia during treatment with atypical antipsychotics should undergo fasting blood glucose testing. In some cases, hyperglycaemia has resolved when the atypical antipsychotic was discontinued; however, some patients required continuation of anti-diabetic treatment despite discontinuation of the suspect drug.

Renal insufficiency

Amisulpride is eliminated by the renal route. In cases of renal insufficiency, the dose should be decreased or intermittent treatment prescribed (see section 4.2).

Epilepsy

Amisulpride may lower the seizure threshold. Therefore patients with a history of epilepsy should be closely monitored during Aktiprol therapy.

Elderly

In elderly patients, amisulpride, like other neuroleptics, should be used with particular caution because of a possible risk of hypotension and sedation. Reduction in dosage may also be required because of renal insufficiency.

Parkinson's disease

As with other antidopaminergic agents, caution should be exercised when prescribing Aktiprol to patients with Parkinson's disease since it may cause worsening of the disease. Aktiprol should be used only if neuroleptic treatment cannot be avoided.

Withdrawal

Acute withdrawal symptoms including nausea, vomiting and insomnia have been described after abrupt cessation of high doses of antipsychotic drugs. Recurrence of psychotic symptoms may also occur, and the emergence of involuntary movement disorders (such as akathisia, dystonia and dyskinesia) has been reported. Therefore, gradual withdrawal of amisulpride is advisable.

Prolongation of the QT interval

Caution should be exercised when amisulpride is prescribed in patients with known cardiovascular disease or family history of QT prolongation and concomitant use with neuroleptics should be avoided. Amisulpride induces a dose-dependent prolongation of the QT interval (see section 4.8). This effect is known to increase the risk of serious ventricular arrhythmias as torsades de pointes. Prior to administration, and if the patient's clinical condition allows, it is recommended to monitor factors that may increase the risk of occurrence of these rhythm disturbances, e.g .

- Bradycardia below 55 bpm
- Electrolyte imbalance, particularly hypokalemia
- Congenital prolongation of the QT interval.
- Existing medications that can cause severe bradycardia (< 55 bpm), hypokalaemia, decreased intracardiac conduction, or prolongation of the QT interval (see section 4.5).

Stroke

In randomized clinical trials versus placebo performed in a population of elderly patients with dementia and treated with certain atypical antipsychotic drugs, a 3-fold increase of the risk of cerebrovascular events has been observed. The mechanism of such risk increase is not known. An increase in the risk with other antipsychotic drugs, or other populations of patients cannot be excluded. Aktiprol should be used with caution in patients with stroke risk factors.

Elderly patients with dementia:

Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Analyses of seventeen placebo-controlled trials (modal duration of 10 weeks), largely in patients taking atypical antipsychotic drugs, revealed a risk of death in drug-treated patients of between 1.6 to 1.7 times the risk of death in placebo-treated patients. Over the course of a typical 10-week controlled trial, the rate of death in drug-treated patients was about 4.5%, compared to a rate of about 2.6% in the placebo group. Although the causes of death in clinical trials with atypical antipsychotics were varied, most of the deaths appeared to be either cardiovascular (e.g., heart failure, sudden death) or infectious (e.g., pneumonia) in nature. Observational studies suggest that, similar to atypical antipsychotic drugs, treatment with conventional antipsychotic drugs may increase mortality.

The extent to which the findings of increased mortality in observational studies may be attributed to the antipsychotic drug as opposed to some characteristic(s) of the patients is not clear.

Aktiprol is not licensed for the treatment of dementia-related behavioural disturbances.

Venous thromboembolism:

Cases of venous thromboembolism (VTE) have been reported with antipsychotic drugs. Since patients treated with antipsychotics often present with acquired risk factors for VTE, all possible risk factors for VTE should be identified before and during treatment with Aktiprol and preventive measures undertaken.

Leukopenia, neutropenia and agranulocytosis

Leukopenia, neutropenia and agranulocytosis have been reported with antipsychotics, including amisulpride. Unexplained infections or fever may be evidence of blood dyscrasia (see section 4.8), and requires immediate haematological investigation.

Breast cancer

Amisulpride may increase prolactin levels and therefore caution should be exercised. Patients with a history of breast cancer or cases of breast cancer in the family should be monitored closely during treatment with Amisulpride.

Benign pituitary tumour

Amisulpride may increase prolactin levels. Cases of benign pituitary tumours such as prolactinoma have been observed during amisulpride therapy (see section 4.8). In case of very high levels of prolactin or clinical signs of pituitary tumour (such as visual field defect and headache), pituitary imaging should be performed. If the diagnosis of pituitary tumour is confirmed, the treatment with amisulpride must be stopped (see section 4.3).

Hepatic toxicity

Severe liver toxicity has been reported with amisulpride use. Patients should be instructed to report immediately signs such as asthenia, anorexia, nausea, vomiting, abdominal pain or icterus to a physician. Investigations including clinical examination and biological assessment of liver function should be undertaken immediately (see section 4.8).

Lactose

This medicinal product contains lactose monohydrate. Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucose-galactose malabsorption should not take this medicine.

4.5 Interaction with other medicinal products and other forms of interaction

• *Contraindicated combinations*

Levodopa: reciprocal antagonism of effects between levodopa and neuroleptics.

Amisulpride may oppose the effect of dopamine agonists e.g. bromocriptine, ropinirole.

Drugs that may induce torsades de pointes or prolong the QT interval (see sections 4.4 and 4.3):

- Class Ia antiarrhythmics such as quinidine and disopyramide.
- Class III antiarrhythmic drugs such as amiodarone and sotalol.
- Other medications such as bepridil, cisapride, sultopride, thioridazine, methadone, erythromycin IV, vincamin IV, halofantrine, pentamidine, sparfloxacin (see section 4.3).

• *Combinations not recommended*

Amisulpride may enhance the central effects of alcohol.

Drugs that increase the risk of torsades de pointes or may prolong the QT interval:

- Drugs that may cause bradycardia, including beta- blockers, bradycardia - inducing calcium antagonists as diltiazem, verapamil, clonidine, guafacin and digoxin.
- Drugs that induce electrolyte imbalance: hypokalaemic diuretics, stimulant laxatives, IV amphotericin B, glucocorticoids and tetracosactides. Hypokalemia should be corrected.
- Antipsychotic drugs as pimozide and haloperidol, imipramine antidepressants, lithium.

• *Combinations to be taken into account*

CNS depressants including narcotics, anaesthetics, analgesics, sedative H1 antihistamines, barbiturates, benzodiazepines and other anxiolytic drugs, clonidine and derivatives

Antihypertensive drugs and other hypotensive medications

Caution is advised when prescribing amisulpride with some antimalarials (e.g., mefloquine) (see section 4.4).

Concomitant treatment with amisulpride and clozapine may result in elevated plasma concentrations of amisulpride.

4.6 Fertility, pregnancy and lactation

Pregnancy

Studies in animals have shown reproductive toxicity.

There are only limited data available from the use of amisulpride in pregnant women. The safety of amisulpride during human pregnancy has not been established.

Amisulpride crosses the placenta.

Use of the drug is not recommended during pregnancy and in women of childbearing potential not using effective contraception, unless the benefits justify the potential risks.

Neonates exposed to antipsychotic drugs during the third trimester of pregnancy are at risk for extrapyramidal and/or withdrawal symptoms following delivery. There have been reports of agitation, hypertonia, hypotonia, tremor, somnolence, respiratory distress, and feeding disorder in these neonates. These complications have varied in severity; while in some cases symptoms have been self-limited, in other cases neonates have required intensive care unit support and prolonged hospitalisation.

Aktiprol should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

Breast-feeding

Amisulpride is excreted into breastmilk in rather large amounts above the accepted value of 10% of the maternal weight-adjusted dosage in some cases, but blood concentrations in breastfed infants have not been evaluated. There is insufficient information on the effects of amisulpride in new-borns/infants. A decision must be made whether to discontinue breast-feeding or to abstain from amisulpride therapy taking into account the benefit of breastfeeding for the child and the benefit of therapy for the woman.

Fertility

A decrease in fertility linked to the pharmacological effects of the drug (prolactin-mediated effect) was observed in treated animals.

4.7 Effects on ability to drive and use machines

Even used as recommended, amisulpride may cause somnolence and blurred vision so that the ability to drive vehicles or operate machinery can be impaired (see section 4.8).

4.8 Undesirable effects

Adverse effects have been ranked under headings of frequency using the following convention:

- very common ($\geq 1/10$);
- common ($\geq 1/100$ to $< 1/10$);
- uncommon ($\geq 1/1,000$ to $< 1/100$);
- rare ($\geq 1/10,000$ to $< 1/1,000$);
- very rare ($< 1/10,000$);
- frequency not known (cannot be estimated from the available data).

Blood and lymphatic system disorders:

Uncommon: Leukopenia, neutropenia (see section 4.4)

Rare: agranulocytosis (see section 4.4)

Immune system disorders:

Uncommon: allergic reaction.

Endocrine disorders:

Common: amisulpride causes an increase in plasma prolactin levels which is reversible after drug discontinuation. This may result in galactorrhoea, amenorrhoea, gynaecomastia, breast pain, and erectile dysfunction.

Rare: benign pituitary tumour such as prolactinoma (see sections 4.3 and 4.4).

Metabolism and nutrition disorders:

Uncommon: hyperglycaemia (see section 4.4), hypertriglyceridemia and hypercholesterolaemia

Rare: hyponatraemia, syndrome of inappropriate antidiuretic hormone secretion (SIADH).

Psychiatric disorders:

Common: insomnia, anxiety, agitation, orgasmic dysfunction

Uncommon: confusion.

Nervous system disorders:

Very common: extrapyramidal symptoms may occur: tremor, rigidity, hypokinesia, hypersalivation, akathisia, dyskinesia. These symptoms are generally mild at optimal dosages and partially reversible without discontinuation of amisulpride upon administration of antiparkinsonian medication. The incidence of extrapyramidal symptoms which is dose related, remains very low in the treatment of patients with predominantly negative symptoms with doses of 50 – 300 mg/day.

Common: somnolence, acute dystonia (spasm torticollis, oculogyric crisis, trismus) may appear. This is reversible without discontinuation of amisulpride upon treatment with an antiparkinsonian agent.

Uncommon: seizures, tardive dyskinesia characterized by rhythmic, involuntary movements primarily of the tongue and/or face have been reported, usually after long term administration. Antiparkinsonian medication is ineffective or may induce aggravation of the symptoms.

Rare: Neuroleptic Malignant Syndrome (see section 4.4), which is a potentially fatal complication.

Not known: restless legs syndrome.

Eye disorders:

Common: blurred vision (see section 4.7).

Cardiac disorders:

Uncommon: bradycardia

Rare: QT interval prolongation, ventricular arrhythmias such as torsade de pointes, ventricular tachycardia, ventricular fibrillation, cardiac arrest, sudden death (see section 4.4).

Vascular disorders:

Common: hypotension

Uncommon: increase in blood pressure

Rare: venous thromboembolism, including pulmonary embolism, sometimes fatal, and deep vein thrombosis (see section 4.4).

Respiratory, thoracic and mediastinal disorders:

Uncommon: nasal congestion, pneumonia aspiration (mainly in association with other antipsychotics and CNS depressants).

Not known: Sleep apnoea*

Gastrointestinal disorders

Common: constipation, nausea, vomiting, dry mouth.

Hepatobiliary disorders:

Uncommon: hepatocellular injury.

Skin and subcutaneous tissue disorders:

Rare: Angioedema, urticaria

Not known: photosensitivity reaction.

Musculoskeletal and connective tissue disorders:

Uncommon: osteopenia, osteoporosis.

Renal and urinary disorders:

Uncommon: urinary retention.

Pregnancy, puerperium and perinatal conditions:

Not known: Drug withdrawal syndrome neonatal (see section 4.6)

Investigations:

Common: weight gain

Uncommon: elevations of hepatic enzymes, mainly transaminases.

*Atypical antipsychotic drugs, such as amisulpride, have been associated with cases of sleep apnoea, with or without concomitant weight gain. In patients who have a history of or are at risk for sleep apnoea, Aktiprol should be prescribed with caution.

4.9 Overdose

Symptoms

Experience with amisulpride in overdosage is limited. Exaggeration of the known pharmacological effects of the drug has been reported. These include drowsiness and sedation, coma, hypotension and extrapyramidal symptoms. Fatal outcomes have been reported mainly in combination with other psychotropic agents.

Management

In cases of acute overdosage, the possibility of multiple drug intake should be considered.

Since amisulpride is weakly dialysed, hemodialysis is of no use to eliminate the drug.

There is no specific antidote to amisulpride.

Appropriate supportive measures should therefore be instituted: close supervision of vital functions and continuous cardiac monitoring (risk of prolongation of QT interval) until the patient recovers.

If severe extrapyramidal symptoms occur, anticholinergic agents should be administered.

Patients where overdose is suspected should be ECG monitored.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Psycholeptics, antipsychotics, benzamides

ATC code N05A L05

Amisulpride binds selectively with a high affinity to human dopaminergic D₂/D₃ receptor subtypes whereas it is devoid of affinity for D₁, D₄ and D₅ receptor subtypes.

Unlike classical and atypical neuroleptics, amisulpride has no affinity for serotonin, -adrenergic, histamine H₁ and cholinergic receptors. In addition, amisulpride does not bind to sigma sites.

In animal studies, at high doses, amisulpride blocks post-synaptic D₂ receptors located in the limbic structures in preference to those in the striatum. Unlike classical neuroleptics it does not induce catalepsy and hypersensitivity of D₂ dopamine receptors does not develop after repeated treatment. At low doses it preferentially blocks pre-synaptic D₂/D₃ receptors, producing dopamine release responsible for its disinhibitory effects.

This atypical pharmacological profile may explain amisulpride's antipsychotic effect at higher doses through post-synaptic dopamine receptor blockade and its efficacy against negative symptoms at lower doses, through pre-synaptic dopamine receptor blockade. In addition, the reduced tendency of amisulpride to produce extrapyramidal side effects may be related to its preferential limbic activity.

5.2 Pharmacokinetic properties

In man, amisulpride shows two absorption peaks: one which is attained rapidly, one hour post-dose and a second between 3 and 4 hours after administration. Corresponding plasma concentrations are 39 ± 3 and 54 ± 4 ng/ml after a 50 mg dose.

The volume of distribution is 5.8 l/kg, plasma protein binding is low (16%) and no drug interactions are suspected.

Absolute bioavailability is 48%.

Amisulpride is weakly metabolised: two inactive metabolites, accounting for approximately 4% of the dose, have been identified. There is no accumulation of amisulpride and its pharmacokinetics remain unchanged after the administration of repeated doses. The elimination half-life of amisulpride is approximately 12 hours after an oral dose.

Amisulpride is eliminated unchanged in the urine. Fifty percent of an intravenous dose is excreted via the urine, of which 90% is eliminated in the first 24 hours. Renal clearance is in the order of 20 l/h or 330 ml/min.

A carbohydrate rich meal (containing 68% fluids) significantly decreases the AUCs, T_{max} and C_{max} of amisulpride but no changes were seen after a high fat meal. However, the significance of these findings in routine clinical use is not known.

Hepatic insufficiency:

Since the drug is weakly metabolised a dosage reduction should not be necessary in patients with hepatic insufficiency.

Renal insufficiency:

The elimination half-life is unchanged in patients with renal insufficiency while systemic clearance is reduced by a factor of 2.5 to 3. The AUC of amisulpride in mild renal failure increased two fold and almost tenfold in moderate renal failure (see section 4.2). Experience is however limited and there is no data with doses greater than 50 mg.

Amisulpride is very weakly dialysed.

Elderly patients:

Limited pharmacokinetic data in elderly subjects (> 65 years) show that a 10-30 % rise occurs in C_{max}, T_½ and AUC after a single oral dose of 50 mg. No data are available after repeat dosing.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Lactose monohydrate 230 mesh, Sodium starch glycolate type A, Hypromellose 2910 E5, Microcrystalline cellulose 101, Magnesium stearate

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

36 months

6.4 Special precautions for storage

Store below 30°C in the original package

6.5 Nature and contents of container

Amisulpride tablets are packed in PVC/PVDC-Alu blisters containing 30, 60 or 100 tablets per box.

400mg: Each blister contains 5 tablets. The respective number of blisters are included in the outer box depending on the pack size.

7. MANUFACTURER

MEDOCHEMIE LTD. (Central Factory), 1-10 Constantinoupoleos street, 3011 Limassol, Cyprus

8. PRODUCT REGISTRATION HOLDER

KOMEDIC SDN BHD, 4 Jalan PJS 11/14, Bandar Sunway, 46150 Petaling Jaya, Selangor

9. DATE OF REVISION OF THE TEXT

12/2024